

TASK-5

EXPLORATORY DATA ANALYSIS (EDA) ON TITANIC DATASET FINDINGS

The objective of this analysis is to explore the Titanic dataset, identify patterns and trends, visualize the data, and gain insights into the factors affecting passenger survival.

During this exploratory data analysis, I analyzed the Titanic dataset to understand the passenger information and identify the factors that influenced survival. The dataset contained 891 passenger records, and the missing values in the Age column were handled by replacing them with the median. No duplicate records were found.

From the visualizations, I observed that most passengers were between 20 and 40 years old and that most of them traveled in Third Class. The fare distribution showed that while most passengers paid lower fares, a few passengers paid very high fares.

The analysis also showed that female passengers had a higher survival rate than male passengers, and passengers traveling in First Class were more likely to survive than those in Third Class. The correlation heatmap indicated that Passenger Class, Fare, and Sex had the strongest relationship with survival, while Age had only a weak relationship.

Overall, this EDA helped me understand the dataset better by identifying patterns, trends, and relationships through statistical analysis and visualizations.